

# Momentum coupling in the “CRIT II” critical ionization velocity experiment

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**Abstract.** A unified theory has been developed to explain the formation of a quasi-dc electric pulse ( $\omega \leq \Omega_i$ ) induced by an ionizing neutral barium beam across an ionospheric plasma. We obtained a generalized form for the dc electric field in the plane perpendicular to the magnetic field within a simplified slab barium cloud moving perpendicular to the geomagnetic fields. A current system associated with the quasi-dc electric field was also proposed to provide a way to transfer the momentum between the streaming barium cloud and the ambient plasma. The characteristic time derived by using this model for momentum coupling between the streaming barium cloud and the ambient plasma is found to be in agreement with the previous result. The quasi-dc electric field predicted by this model is reasonably consistent with the “CRIT II” critical ionization velocity measurements. On the basis of the constraints of the conservation of energy and momentum, we found that Alfvén’s critical ionization velocity (CIV) effect is a self-limiting ionization process in a finite extent neutral cloud. It may be the reason why the CIV effect took place in CRIT II but lasted only for a very short period, and it may have resulted in low ionization yields in most of space CIV experiments.

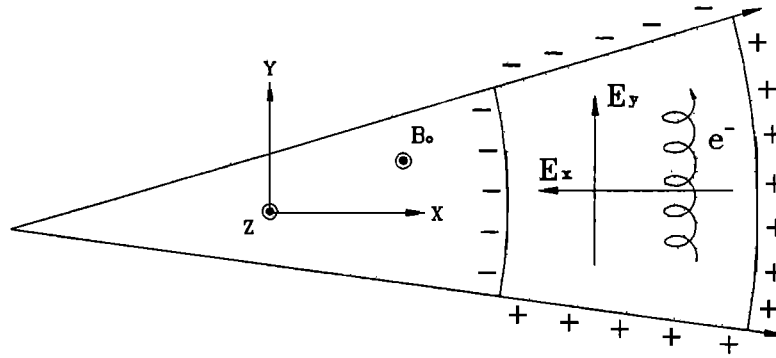
## Introduction

Upper ionospheric barium release using sounding rockets to test Alfvén’s critical ionization velocity hypothesis (1954) has been conducted actively more than a half dozen times over the last decade [e.g., Haerendel, 1982; Wescott *et al.*, 1986; Wescott *et al.*, 1990; Stenbaek-Nielsen *et al.*, 1990a, b; Torbert *et al.*, 1992]. According to Alfvén’s critical ionization velocity (CIV) hypothesis, a rapid gas ionization will occur when a neutral gas cloud moves across a magnetized plasma with a relative velocity perpendicular to the magnetic field exceeding a critical value,  $v_c$ , defined by  $\sqrt{2e\phi_{\text{ion}}/M}$ , where  $\phi_{\text{ion}}$  and  $M$  are the ionization potential and mass of neutral atom, respectively. In a typical shaped-charge barium release experiment a directed neutral beam of heavy atoms, usually barium, with 50% of populations exceeding its critical ionization velocity is injected into an ionospheric plasma. Therefore one would expect a substantial amount of the barium atoms will be ionized by the CIV effect. As reviewed by Torbert [1990],

most of space CIV experiments have yielded negative results except for the project Porcupine [Haerendel, 1982] (produced 10–20% ionization yield) and CRIT II (Critical Ionization Test II) [Stenbaek-Nielsen *et al.*, 1990b] (produced 2–4% ionization yield). Although laboratory experiments have repeatedly confirmed the existence of the CIV effect [Fahleson, 1961; Himmel *et al.*, 1977; Axnäs, 1978] and a number of theoretical studies [Raadu, 1978; Formisano *et al.*, 1982; Galeev and Chabibrachmanov, 1983; Machida and Goertz, 1986; Möbius *et al.*, 1987] have shown relatively consistent theories to explain these experiments, the conflicting results yielded from space CIV experiments have posed a question as to the existence of the CIV effect in a free space where it was proposed to have taken place.

One of the most interesting phenomena arising from the motion of an ionizing barium cloud in the upper atmosphere is the formation of a self-polarization electric field within the cloud. Considering an ionizing neutral beam of finite extent moving across a magnetic field  $B_0$  (along the  $z$  axis) in the  $x$  direction (see Figure 1), there will be a Hall current,  $j_y$ , which leads to charge accumulation on the boundary of the beam creating a transverse electric field,  $E_y$ . This electric field will consequently cause the ambient plasma undergoing an  $E \times B$  drift across the magnetic field to reduce the relative velocity between the neutral beam and the ambient plasma. If the process of accumulation of charges

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**Figure 1.** Polarization electric fields within a finite extent ionizing neutral beam.

continues, this polarizing electric field can grow substantially and finally shut down the CIV effect. Moreover, the newly created massive barium ions in the beam cloud will leave their electron pairs behind and create a longitudinal electric field,  $E_x$ , which is antiparallel to the beam direction at the leading edge of the ionizing neutral cloud. According to the ionizing front model [Piel *et al.*, 1980], this longitudinal electric field is responsible for electrons performing the  $E \times B$  drift in the direction transverse to the ambient magnetic field and driving the modified two stream instabilities to sustain the CIV effect. This type of quasi-dc electric field associated with a finite extent barium beam has been observed in a couple of space CIV experiments [Kelley *et al.*, 1991; Swenson *et al.*, 1992]. It is thought that the quasi-dc electric field may have played an important role in the momentum transfer in space CIV experiments. In order for a CIV process to survive, electrons must absorb enough energy from beam ions to ionize more neutrals. Since electrons are massless and in order to conserve momentum, the ambient plasmas must absorb the momentum corresponding to the amount of energy being transferred from the beam ions to electrons. The existence of quasi-dc electric fields implies a current system with which the momentum balance within an ionizing neutral cloud can be managed. Therefore it is important to understand the electric fields inside the cloud, and it could be one of the keys to uncovering the mystery of the observed highly variable ionization yields in ionospheric experiments.

The purpose of this study is to provide a theoretical foundation for the formation of a quasi-dc electric field for CIV experiments involving shaped-charge beam releases. In section 2, we will review two theoretical models which pay attention to the formation of the quasi-dc electric fields associated with a streaming neutral cloud. Section 3 describes the basic assumptions and physics for the model. In section 4, we test our model results with the CRIT II measurements, and finally a conclusion will be given in section 5.

## Review of the Two Previous Models

The polarization electric field has been derived by the so-called “momentum coupling model” [Haerendel,

1982]. Haerendel proposed that a dynamo current,  $J_D$ , induced by the injection of an ionizing neutral beam inside a magnetic flux tube will be closed by a polarization current,  $J_P$ , associated with emitting Alfvén waves. The exchange of momentum between the newly created ions in the beam cloud and the ambient plasma is explained by Haerendel as follows. The dynamo current will slow the newly created ions by a  $J_D \times B_0$  force, while the polarization current, in the opposite direction of the dynamo current, within the Alfvén wave fronts will accelerate the ambient plasma through a  $J_P \times B_0$  force in the direction of the neutral beam, thus absorbing the momentum. Applying the conservation of momentum to a flux tube, a transverse electric field (with respect to the beam velocity) which is characterized by the mass loading factor,  $\lambda_M$ , is obtained and is given by

$$\vec{E} = -\frac{\lambda_M}{1 + \lambda_M} \frac{\vec{V}_b \times \vec{B}_0}{c}, \quad (1)$$

where

$$\lambda_M = \frac{M_b \dot{n}_i L_{\parallel}}{2\rho_a v_A}, \quad (2)$$

$\vec{V}_b$  and  $M_b$  are the velocity and the mass of the neutral beam,  $\rho_a$  is the mass density of the ambient plasma,  $v_A$  is the Alfvén speed,  $\dot{n}_i$  is the ionization rate of the neutral beam, and  $L_{\parallel}$  is the beam width along the geomagnetic field. It should be noted that the mass loading factor is proportional to the ionization rate but inversely proportional to the squared root of the ambient plasma density.

This momentum coupling model does not include the charge separation in the direction of beam. Therefore it cannot explain the longitudinal component of electric field (parallel to the neutral beam velocity) which has also been observed in a couple of space CIV experiments [Kelley *et al.*, 1991; Swenson *et al.*, 1992]. An identical result derived from a totally different approach was also obtained by Torbert [1988]. The basic idea of Torbert’s model is to solve the closure of a current system within a beam cloud in the rest frame of neutrals.

Another model of interest is the “barium swarm” [Brenning *et al.*, 1991b]. The basic idea of this model is that a generator current system induced by the ion

motion across the magnetic field will be closed by field-aligned currents associated with two Alfvén waves launching from two sides of the barium cloud. An electric field within a simplified cylinder cloud of barium was maintained by this current system. A quasi-steady state solution for the electric field was given by this model as

$$\vec{E} = -\frac{K}{1+K^2} \frac{B_0 \vec{V}_{b\perp}}{c} - \frac{K^2}{1+K^2} \frac{\vec{V}_b \times \vec{B}_0}{c}, \quad (3)$$

where

$$K = \frac{\Delta n e v_A \pi L_{\parallel}}{B_0 c}, \quad (4)$$

$\Delta n$  is the deposited ions due to ionization. This quasi-steady electric field is entirely determined by the  $K$  parameter, which is proportional to the deposited ions and reciprocally proportional to the squared root of the ambient plasma density.

Although the barium swarm model can predict both longitudinal and transverse electric fields inside a barium cloud and does a better match on the electric fields for the CRIT experiments, it is not suitable for a streaming cloud with a low deposited ion density and a high ionization rate. On the other hand, the momentum coupling model is not appropriate for the case when the ionization rate is low and the deposited ion density is high. In the present paper, we will use a different approach but one similar to that of *Torbert* [1988] to further study this quasi-dc electric field inside an ionizing neutral cloud. The deposited ion density and the ionization rate will be considered to obtain a generalized form for the quasi-dc electric fields and, most important, to explain the CRIT II measurements.

## Model Description

In this model, a homogeneous neutral barium cloud is injected into an ionospheric plasma. All types of collisions will be ignored as well as the expansion of the neutral cloud. We will consider time variations on a timescale slower than the ambient plasma ion gyro period. The beam velocity,  $v_b$ , is chosen to be in the  $x$  direction and the ambient magnetic field,  $B_0$ , in the  $z$  direction. The ionizing neutral cloud is highly simplified as a rectangular shape with a characteristic length,  $L_{\parallel}$ , along the magnetic field line and a cross section area,  $L_x \times L_y$ , in the plane perpendicular to  $B_0$ . The frame of reference is considered to be in the rest frame of the neutral beam. Therefore all the neutral barium atoms are initially stationary, and the ambient plasma will be moving at the speed of neutral beam but in the  $-x$  direction. This would imply an electric field,  $E_0 = V_b B_0 / c$ , in the  $-y$  direction due to the motion of plasmas across the ambient magnetic field. A systematic diagram for the model of the formation of quasi-dc electric fields is shown in Figure 2.

In Figure 2a, once a neutral barium atom is ionized (denoted as an asterisk in stage 1), the electric field,  $\vec{E}_0$ , will accelerate the newly created barium ion in the  $-y$  direction and the electron in the opposite direction. Since electrons are fully magnetized and the

heavy barium ions are effectively unmagnetized in the timescale of oxygen gyro period, the newly created ion will leave electron behind and create a current density,  $j_{ba} = \dot{n}_i e \rho_L$ , in average (averaged over one half of a barium gyro period) in the same direction as  $\vec{E}_0$ , where  $\rho_L = Ec / B_0 \Omega_b$ . It should be noted that this value of averaged current density is valid only when the width of the injected cloud is greater than or at least equal to the barium ions gyro radius. If the cloud is small, discontinuities in the current density will occur. The ion-electron separation will result in a slowly oscillating electric field pointing along  $\hat{y}$ , which in turn will reduce the initial electric field inside the cloud from  $E_0$  to  $E_y$ . Meanwhile (Figure 2b, stage 2), a Hall current density,  $j_{hall} = \Delta n e c E_y / B_0$ , will be created inside the neutral cloud in the  $x$  direction due to excess electrons,  $\Delta n$ , performing the  $E_y \times B_0$  drift in the  $-x$  direction. These drifting electrons will be accumulated on the left boundary of the beam to polarize the neutral cloud and create a new electric field  $E_x - \hat{x}$ . Due to this electric field (Figure 2c, stage 3), the ambient plasma will undergo the  $E_x \times B$  drift in the  $y$  direction but will create no current. Only the excess electrons can contribute a Hall current density,  $j_{hall} = \Delta n e c E_x / B_0$ . Again, these Hall current electrons will charge up the top side boundary to create an electric field and reduce the original electric field  $\vec{E}_0$ . If there is no current sink, this process of charge accumulation will continue until the electric field inside the cloud is reduced to zero.

The system current density and electric field within the neutral cloud are given by adding up the total current density and the electric field described above (illustrated in Figure 3). After summing up all the sources of current, one finds a linear relation between the system current density and electric field as follows,

$$j_x = \frac{\Delta n e c E_y}{2 B_0} \quad (5)$$

$$j_y = (1-s) \frac{\dot{n}_i M c^2 E_y}{B_0^2} + \frac{\Delta n e c E_x}{2 B_0}, \quad (6)$$

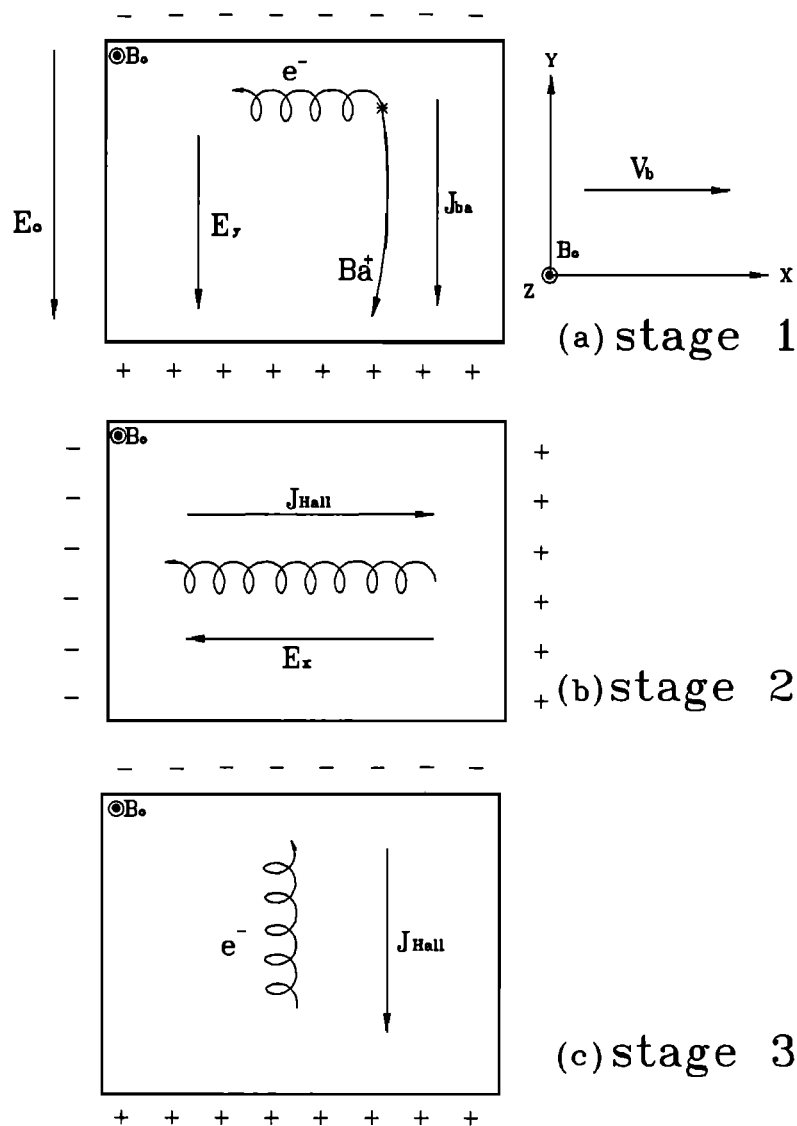
where a parameter,  $s$ , is used to separate two timescale problems. For time variation longer than one ion beam gyro period,  $s$  is equal to 1. Otherwise,  $s$  is equal to 0. In order to conserve the total charge, the Hall current terms are multiplied by a factor of  $1/2$ .

Furthermore, we assume that the current density is uniform within the ionizing neutral cloud. Thus the system current can be obtained by multiplying the current density by the area through which it passes; that is,

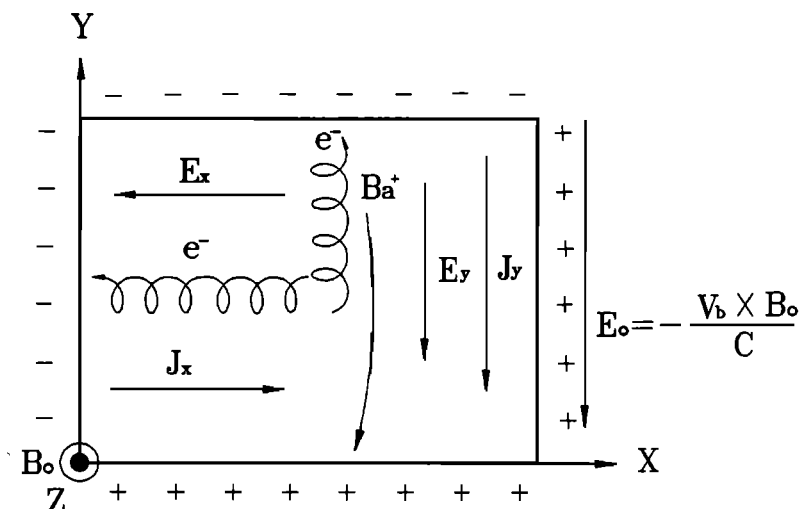
$$I_x = \frac{\Delta n e c E_y}{2 B_0} L_y L_{\parallel} \quad (7)$$

$$I_y = (1-s) \frac{\dot{n}_i M c^2 E_y}{B_0^2} L_x L_{\parallel} + \frac{\Delta n e c E_x}{2 B_0} L_x L_{\parallel}. \quad (8)$$

As mentioned previously, currents will be accumulated on the cloud boundaries. In order to maintain charge neutrality, the ambient plasma must take up



**Figure 2.** Schematic current system and its associated electric fields inside an ionizing cloud in the reference frame of the neutral beam: (a) direct ion current and retarding electric field, (b) electron Hall current induced by retarding electric field, which polarizes the beam cloud, and (c) electron Hall current created by polarization electric field.



**Figure 3.** System currents and electric fields within an ionizing cloud in the neutral beam frame of reference.

excess electrons or supply electrons to neutralize any charge imbalance inside the ionizing neutral cloud. This process can be accomplished by field-aligned currents associated with emitting two shear Alfvén waves from each side of the cloud along the magnetic field line. The field-aligned current can be calculated by combining the Ampere's law and Faraday's law. With neglecting the displacement current and eliminating  $B$ , one obtains

$$\frac{\partial \vec{J}}{\partial t} = -\frac{c^2}{4\pi} \vec{\nabla} \times \vec{\nabla} \times \vec{E}. \quad (9)$$

The  $z$  component, assuming  $\nabla_{\perp}^2 E_z = 0$ , is

$$\frac{\partial j_z}{\partial t} = -\frac{c^2}{4\pi} \frac{\partial}{\partial z} \left( \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} \right). \quad (10)$$

Knowing that two Alfvén waves will be emitted in either direction along the magnetic field line, the combination of the time and spatial derivative can be replaced by the Alfvén speed to give

$$j_z = \Sigma_A \left( \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} \right), \quad (11)$$

where  $\Sigma_A = c^2/4\pi v_A$  is the Alfvén conductivity. Thus

$$\begin{aligned} I_{\parallel} &= \int \int \Sigma_A \left( \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} \right) dx dy \\ &= \Sigma_A L_y E_x + \Sigma_A L_x (E_o - E_y) \\ &= I_{\parallel x} + I_{\parallel y}, \end{aligned} \quad (12)$$

where the subscript  $z$  has been replaced by  $\parallel$  denoting the component parallel to  $B_o$ . The electric field inside the cloud can be derived from an analogy of parallel charged plates on the  $x$  and  $y$  boundaries of the beam in a dielectric media. One has

$$\epsilon E_x = 4\pi \frac{Q_x}{L_y L_{\parallel}} \quad (13)$$

$$\epsilon E_y = E_o - 4\pi \frac{Q_y}{L_x L_{\parallel}}, \quad (14)$$

where  $\epsilon = 1 + \omega_a^2/\Omega_a^2 + s\omega_b^2/\Omega_b^2$  is the low-frequency dielectric constant of the ionospheric plasma,  $\omega_i(\Omega_i)$  is the plasma (gyro) frequency of species  $i$ , the subscript  $a(b)$  denotes the ambient plasma (beam), and  $Q_x$  and  $Q_y$  are the total charges on the bottom and the left plates, respectively. In Figure 3, we have also assumed that an equal amount of negative charges was built up on the top and bottom plates. The surface charges can be eliminated by applying the conservation of the total charges to the system to give

$$\begin{aligned} \dot{Q}_x &= I_x - 2I_{\parallel x} \\ &= \frac{\Delta nec L_y L_{\parallel}}{2B_o} E_y - 2\alpha L_y \Sigma_A E_x \end{aligned} \quad (15)$$

$$\begin{aligned} \dot{Q}_y &= I_y - 2I_{\parallel y} \\ &= (1-s) \frac{\dot{n}_i M c^2 L_x L_{\parallel}}{B_o^2} E_y + \frac{\Delta nec L_x L_{\parallel}}{2B_o} E_x \\ &\quad - 2\alpha L_x \Sigma_A (E_o - E_y), \end{aligned} \quad (16)$$

in which a factor of 2 on the field-aligned currents is due to the fact that two Alfvén waves will be emitted and a current-limiting factor,  $\alpha$ , is introduced in order to consider the maximum current that the ionospheric plasma can supply. This maximum current, however, is unknown. It may relate to the critical current for electrostatic ion cyclotron instability proposed by *Lysak and Dum* [1983]. If the parallel current is smaller than the maximum current, then  $\alpha = 1$ ; otherwise,  $\alpha < 1$ . A system equation for the time dependent electric field inside a neutral cloud can be obtained by taking time derivative to (13) and (14) and combining it with (15) and (16) to delete the charge terms. With some algebraic manipulations, one has

$$\gamma \dot{E}_x = -(1 + s\lambda_M) E_x + \kappa E_y \quad (17)$$

$$\gamma \dot{E}_y = -\kappa E_x - (1 + \lambda_M) E_y + E_o, \quad (18)$$

where  $\gamma = L_{\parallel} \epsilon / 8\pi \alpha \Sigma_A$ ,  $\kappa (= L_{\parallel} \Delta nec / 4B_o \alpha \Sigma_A)$  is identical to the barium swarm's  $K$  factor [Brenning *et al.*, 1991b] when  $\alpha = 1$  and  $\lambda_M (= \dot{n}_i M c^2 L_{\parallel} / 2B_o^2 \alpha \Sigma_A)$  is identical to the mass loading factor [Haerendel, 1982; Torbert, 1988] when  $\alpha = 1$ . A quasi-steady solution can be obtained algebraically by setting the time derivatives on the left hand side of (17) and (18) to be zero. One has

$$E_{0x} = \frac{\kappa}{(1 + \lambda_M)(1 + s\lambda_M) + \kappa^2} E_o \quad (19)$$

$$E_{0y} = \frac{1 + s\lambda_M}{(1 + \lambda_M)(1 + s\lambda_M) + \kappa^2} E_o. \quad (20)$$

After transforming back to the plasma reference frame, the total electric field, in a vector form, becomes

$$\begin{aligned} \vec{E} = & - \frac{\kappa}{(1 + \lambda_M)(1 + s\lambda_M) + \kappa^2} \frac{B_o}{c} \vec{V}_b \\ & - \frac{\lambda_M(1 + s\lambda_M) + \kappa^2}{(1 + \lambda_M)(1 + s\lambda_M) + \kappa^2} \frac{\vec{V}_b \times \vec{B}_o}{c}. \end{aligned} \quad (21)$$

The first term on the right hand side of expression (21) is the longitudinal electric field which is antiparallel to  $\vec{V}_b$  and the second term is the transverse electric field with respect to  $\vec{V}_b$ . The difference between this model result, (21), and others, (1) and (3), is obvious. Expression (21) is essentially a combination of the two results from the "momentum coupling" and the "barium swarm" models. The quasi-dc electric field is now determined by the mass loading factor,  $\lambda_M$ , and the  $\kappa$  parameter. When  $\kappa = 0$  and  $\alpha = 1$ , only the transverse electric field exists and it is identical to the momentum coupling model result, (1). On the other hand, when  $\lambda_M = 0$  and  $\alpha = 1$ , it reduces to the barium swarm model result, (3).

Equations (17) and (18) can be solved if all the related parameters are independent on time. Accordingly, the characteristic time,  $\tau_{DC}$ , the time for setting up a static electric field, is found to be  $\gamma(1 + \lambda_M)^{-1}$ . This  $\gamma$  parameter has a physical meaning associated with momentum balance between the ambient plasma and the beam ions. The plasma dielectric constant can be rewritten as  $\epsilon = 1 + c^2/v_A^2(1 + s\rho_b/\rho_a)$ , where  $\rho_a(\rho_b)$

is the mass density of the ambient plasma (beam). In the ionospheric plasma the Alfvén speed is much less than the speed of light; therefore the vacuum term can be neglected from the dielectric constant. With some reductions, the characteristic time becomes

$$\tau_{DC} = (1 + \lambda_M)^{-1} (L_{\parallel} / 2\alpha v_A) (1 + s\rho_b / \rho_a). \quad (22)$$

When  $\lambda_M$  is small and  $\alpha = 1$ , (22) reduces to

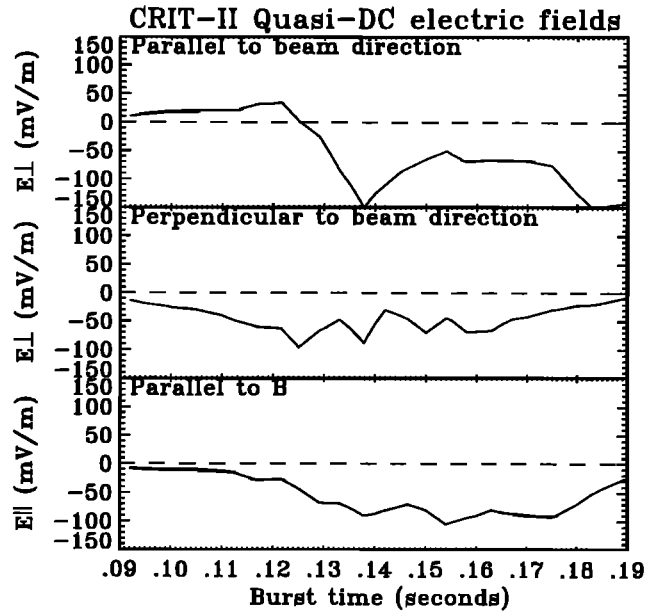
$$\tau_{DC} \sim (L_{\parallel} / 2v_A) (1 + \rho_{ba} / \rho_o) \quad (23)$$

for a long time variation case ( $s = 1$ ). It is consistent with Scholer's [1970] result. The physical meaning of this time constant is that the time required to reach a static electric field (i.e., momentum coupling time) is equivalent to the time needed for Alfvén waves covering an ambient plasma equal to the mass of the ionization cloud. However, if  $\alpha < 1$ , the momentum coupling time will be longer and consequently will slow the ionization down.

## Comparison With CRIT II Data

In order to test our model, we need to know the neutral barium ionization rate (for  $\lambda_M$ ) and the barium ion density (for  $\kappa$ ) in addition to the electric and magnetic fields (to determine  $\alpha$ ). Although the quasi-dc electric fields has also been observed in the CRIT I experiment, only the CRIT II experiment made a successful particle measurement inside a barium beam. Therefore we will use CRIT II data to test this model.

CRIT II was instrumented to make the field and particle measurements inside and outside a barium beam [Torbert *et al.*, 1992]. The observed quasi-dc electric field at the CRIT II main payload during the first release (note that there were two releases conducted in the CRIT II experiment) is shown in Figure 4 [Swenson *et al.*, 1992]. The top panel presents the longitudinal (with respect to the beam direction) electric field,  $E_{\parallel}$ , and the middle one presents the transverse electric field,  $E_{\perp}$ . The geomagnetic field-aligned electric field is shown in the bottom panel. Before 0.12 s (approximately the time right before the main neutral barium beam arrived at the instrumented payload),  $E_{\parallel}$  was rather stable and close to zero, while  $E_{\perp}$  was growing steadily up to its maximum value of 100 mV m<sup>-1</sup> at 0.125 s. The growth of  $E_{\parallel}$  did not happen until 0.12 s. It quickly reached its maximum value of 150 mV m<sup>-1</sup> at 0.138 s though. This feature of growing electric field can be explained by this model result. When  $t \leq 0.12$  s, there were few deposited barium ions, but the barium ionization rate was very high and it increased with increasing the barium neutral density. This will give a small value of  $\kappa$  but a large value of  $\lambda_M$  initially. One would expect that  $E_{\parallel}$  (proportional to  $\Delta n_i$ ) will be small but  $E_{\perp}$  (proportional to  $\dot{n}_i$ ) will be large and be increasing with time. During the passage of the main barium cloud ( $t \sim 0.135$  s), both  $\Delta n_i$  and  $\dot{n}_i$  reached their peak value and consequently  $E_{\parallel}$  and  $E_{\perp}$  reached their maximum value.



**Figure 4.** Quasi-static electric field (0–60 Hz) measurement in magnetic coordinates (after Swenson *et al.* [1992] with kind permission from Elsevier Science Ltd., The Boulevard, Langford Lane, Kidlington OX5 1GB, UK).

The characteristic value of quasi-dc electric fields measured at the CRIT II main payload can be calculated from (21) for a given  $\lambda_M$ ,  $\kappa$ , and  $s$ . To estimate these parameters, we will use the data taken from the onboard CRIT II measurements. The current limiting factor,  $\alpha$ , has been estimated based on the CRIT II data ranging from 0.05 to 0.2 for high-frequency field oscillations in wave growth phase ( $t \leq 0.12$  s) [Swenson *et al.*, 1992]. However,  $\alpha$  for low-frequency waves was not given. Here we will use  $\alpha = 0.1$  as a typical value for the following calculation and discussion. Later, we will see that this is the best choice in order to match the observations. The plasma density was measured to be  $5.41 \times 10^5$  cm<sup>-3</sup>, and the ambient magnetic field was 0.43 G. The spatial extent of the barium beam parallel to the magnetic field was about 500 m for the first release and 1200 m for the second release. In CRIT II, the timescale of the  $E$  field is less than one barium gyro period. Therefore we can take  $s$  to be 0. With  $\Delta n = 1 \times 10^5$  cm<sup>-3</sup> and  $\dot{n}_i = 1 \times 10^6$  cm<sup>-3</sup> s<sup>-1</sup> for the first release [Liou and Torbert, 1995], we obtain  $\lambda_M = 0.12$  and  $\kappa = 0.19$  or  $E_{\parallel} = 70$  mV m<sup>-1</sup> and  $E_{\perp} = 58$  mV m<sup>-1</sup> for the first release. These values differ from the observations by a factor of 2. It should be noted that the quasi-dc electric field shown in Figure 4 was low passed at 60 Hz. This model is suitable for a time variation of the same order of the barium gyrofrequency ( $\sim 5$  Hz). If this factor is considered, this difference should be reduced. One may ask how was the quasi-dc electric field ( $f \leq 60$  Hz) shown in Figure 4 created? It may be that an oxygen ion beam was created by the “snow-plow” effect as the barium beam swept through the ambient plasma. This oxygen beam

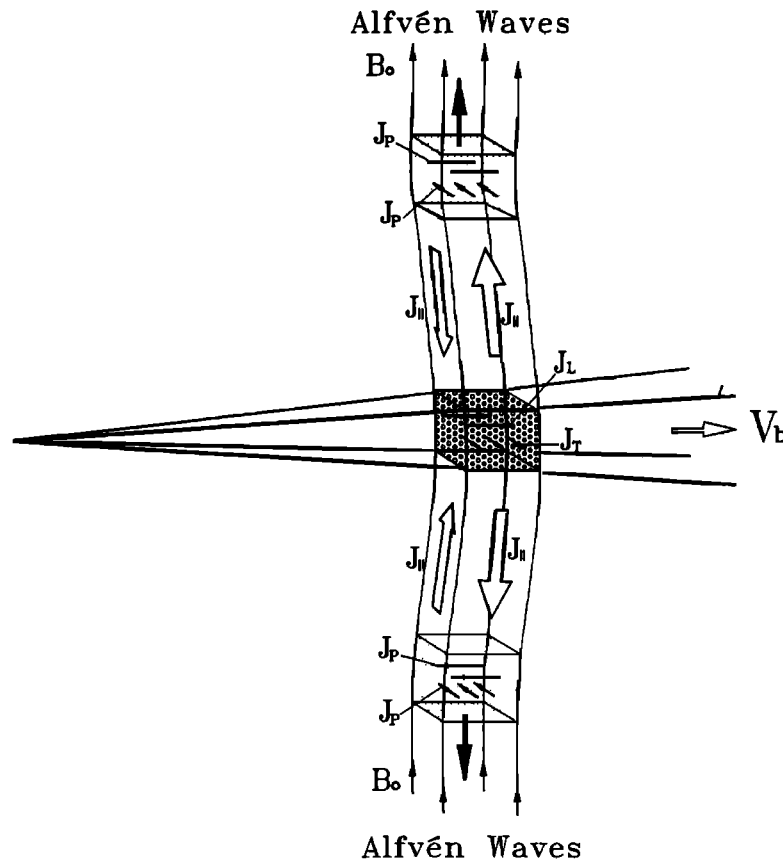
may mimic the newborn ions ahead of and inside the barium beam. The increase of plasma density has been measured by CRIT II and reported by *Swenson et al.* [1992]. With the average plasma density of  $\sim 8 \times 10^5$  to  $9 \times 10^5 \text{ cm}^{-3}$  at 0.12–0.14 s, which corresponds to  $\Delta n \sim 3 \times 10^5 \text{ cm}^{-3}$ , one finds from our model that  $\kappa = 0.57$  or  $E_l = 170 \text{ mV m}^{-1}$  and  $E_t = 131 \text{ mV m}^{-1}$ . It is consistent with the observations. For the second release, however, the electric field data have not been fully analyzed and published at this time.

In addition to the transverse quasi-dc electric, the geomagnetic field-aligned electric field has also been observed on CRIT I [*Kelley et al.*, 1991] and CRIT II [*Swenson et al.*, 1992]. However, this model fails to explain the field-aligned dc electric field. A possible reason for the creation of the parallel dc electric field has been proposed by *Brenning et al.* [1991a, b]. They suggested that the field-aligned current given by (11) could exceed the maximum current which can be supplied by the ionospheric plasma (i.e.,  $\alpha < 1$ ); therefore charge imbalance inside the barium cloud cannot be neutralized. However, it may be that since electrons cannot freely leave ions behind due to the electrostatic drags, as CIV is taking place, heated electrons will tend to escape from the ionizing barium cloud and create a self-contained electric field. The nominal escaping speed for electrons in a plasma is about the ion acoustic speed [*Singh*, 1990]. If ambipolar diffusion occurs, a polar-

ization electric field will be created along the magnetic field. The polarization electric field is estimated to be about  $(kT_e/q)(\nabla n/n)$ , where  $\nabla = \partial/\partial z$ . The typical energy for the heated electrons on CRIT II was about 10 eV. Here  $\nabla n$  was of the order of  $n$ , and the characteristic length scale of the inhomogeneity was about 200 m. This leads to a field-aligned electric field of  $\sim 50 \text{ mV m}^{-1}$ , which is reasonably consistent with the measurements.

## Discussion

To conserve the momentum in a critical ionization velocity interaction, a polarization electric field must develop as we mentioned previously. In previous section, we have developed a model for the quasi-dc electric field observed on CRIT II. On the basis of this dc electric field model, a possible current system inside and outside a streaming ion cloud in a magnetic flux tube can be drawn and shown in Figure 5. The momentum balance between the ion cloud and the ambient plasma can be managed by this current system. Within the cloud a transverse current  $J_T$  will slow the ion beam through a  $J_T \times B_0$  force. The momentum loss from the ions is delivered to the ambient plasma by Alfvén waves through a  $J_P \times B_0$  force within the wave fronts. Likewise, the momentum balance in the transverse direction can be managed by the longitudinal current  $J_L$  within



**Figure 5.** An approximate current system inside and outside an ionizing neutral beam along a magnetic flux tube.

the beam cloud and the wave polarization current  $J_P$  within the wave fronts through  $J \times B_0$  forces.

From the energy point of view, the condition to sustain the CIV ionization effect is

$$\eta_e \frac{1}{2} M \left( \vec{v}_{b\perp} - c \frac{\vec{E} \times \vec{B}_0}{B_0^2} \right)^2 \geq (1 + \theta) e \phi_{ion} \quad (24)$$

where  $\eta_e$  is the fraction of ion beam energy being converted to electron energy,  $\theta$  is the energy ratio of the excitation loss to the ionization loss in electron impact collisions, and  $\phi_{ion}$  is the ionization potential for the beam atoms. For a barium beam release,  $\theta$  is equal to 1 because in electron impact collisions, about 50% of hot electron energies will be wasted to the barium line excitation [Newell and Torbert, 1985]. Therefore a double amount of ionization energy is needed in average to ionize one barium atom. Substituting (21) into (24), the threshold velocity for CIV to occur is

$$v_{thr} = \sqrt{\frac{1 + \theta}{\eta_e} \frac{(1 + \lambda_M)(1 + s\lambda_M) + \kappa^2}{\sqrt{(1 + s\lambda_M)^2 + \kappa^2}}} v_c. \quad (25)$$

The physical implications resulting from this threshold condition is obvious. Let's consider an initial stage of CIV, i.e.,  $\lambda_M \sim 0$ ,  $\kappa \sim 0$ . One can see that the threshold velocity for CIV to occur is  $\sqrt{(1 + \theta)/\eta_e} v_c$ , which is always greater than the Alfvén's critical ionization velocity because of the constraint of energies available for ionization. The factor of  $\sqrt{1 + \theta}$  is obviously due to the beam atoms line excitation, and therefore it will vary with different neutral species. The factor of  $1/\sqrt{\eta_e}$  is due to imperfect energy transfer between beam ions and electrons. In a typical beam release CIV experiment, barium beam velocities can reach as high as  $5v_c$ . With a maximum value of  $2/3$  for  $\eta_e$  [Formisano et al., 1982], one can see that the Alfvén's CIV condition can be easily met. However, after CIV is triggered,  $\lambda_M$  and  $\kappa$  will be increasing and consequently leveling up the threshold velocity. Therefore for a given  $\eta_e$ , CIV will settle at a higher threshold velocity depending on the ionization rate and the density of deposited ions. This conclusion implies that the CIV effect in a finite extent beam is a self-limiting ionization process. In addition, since both  $\lambda_M$  and  $\kappa$  are inversely proportional to the square root of the ambient plasma density, in a higher (lower) density plasma the threshold velocity for CIV decreases (increases).

It is worth noting that according to the ion data shown in Figure 4 of Torbert et al. [1992], a band of energetic barium ions, starting at 0.13 s of burst time, quickly lost their energy and momentum in 50 ms. This feature of anomalous barium energy and momentum lost has been suspected as a strong evidence of the CIV effect [Torbert et al., 1992]. One can see from Figure 4 that the quasi-dc electric field antiparallel to the beam reached a maximum value after  $\sim 0.14$  s. Therefore a strong momentum coupling is expected to happen. However, this narrow feature of momentum lost in time suggests that the CIV effect was quenched right after the strong momentum coupling. This quick

stoppage of CIV may be explained by (25). The  $\eta_e$  parameter is calculated based on the CRIT II data to be in the vicinity of 0.2 at 0.12–0.14 s. Here we will use 0.2 as a characteristic value for discussions. With  $\lambda_M = 0.12$  and  $\kappa = 0.19$ , we obtain  $v_{thr} \sim 3.6v_c$ , which is equivalent to 67 eV of barium energy. Because the energies of barium atoms arrived at the payload after 0.14 s are below 65 eV, one would suspect that CIV may have ceased at that time.

From (25) the maximum bound for the ionization rate and the density increased within a beam cloud will be

$$\Delta n \leq \frac{4B_0 \alpha \Sigma_A}{ecL_{\parallel}} \sqrt{\Gamma^2 - 1} \quad (26)$$

$$\dot{n}_i \leq \frac{2B_0 \alpha \Sigma_A}{ecL_{\parallel}} (\Gamma - 1) \Omega_b, \quad (27)$$

where  $\Gamma = v_b/v_{thro}$  and  $v_{thro} = \sqrt{2/\eta_e} v_c$ .

In CRIT II, the typical velocity of the beam in the direction perpendicular to the ambient magnetic field was about  $3.7 v_c$ . With  $\eta_e = 0.2$  and  $\alpha = 0.1$ , we obtain  $\Delta n \leq 3.3 \times 10^5 \text{ cm}^{-3}$  and  $\dot{n}_i \leq 1.4 \times 10^6 \text{ cm}^{-3} \text{ s}^{-1}$  for the first release and  $\Delta n \leq 1.4 \times 10^5 \text{ cm}^{-3}$  and  $\dot{n}_i \leq 5.8 \times 10^5 \text{ cm}^{-3} \text{ s}^{-1}$  for the second release, which is reasonably consistent with the CRIT II measurements [Liou and Torbert, 1995].

Most of space CIV experiments were conducted at a lower density plasma, about a factor of 20–30 smaller than that in CRIT II. Since  $\alpha$  is proportional to the ambient plasma density,  $n_a$ , and  $\Sigma_A$  is proportional to  $\sqrt{n_a}$ , we found that the maximum bound for the ion density increased by CIV is proportional to  $n_a^{3/2}$ . Therefore we conclude that the ionization yield for low-density plasma experiments will be less than that observed in CRIT II by 2 orders of magnitude, provided there is similar experimental environment. In a beam release experiment, the beam density drops as  $r^{-3}$ . Since  $L_{\parallel} \propto r$ , one would expect that ion production will decrease roughly as  $L_{\parallel}^{-3}$ . However, from (26), we found  $\Delta n \propto L_{\parallel}^{-1}$ . The reason for this proportionality is not clear, but it interprets quite well the results of the two barium beam releases in CRIT II. An interesting result from (26) is that  $\Delta n$  is not dependent on the energy of the released beam. Instead, it depends on the beam velocity through  $\sqrt{(v_b/v_{thro})^2 - 1}$ . At low beam velocity ( $v_b \sim v_{thro}$ ),  $\Delta n \propto \sqrt{v_b - v_{thro}}$ . When  $v_b \gg v_{thro}$ , we found  $\Delta n \propto v_b$ .

## Conclusions

We have presented a qualitative model based on the closure of a current system within an ionizing neutral cloud to investigate the formation of quasi-dc electric fields inside a streaming neutral cloud. Despite the simplified nature of the model, a generalized result, which is essentially a combination of the results from the two previous theoretical models “momentum coupling” and “barium swarm,” is obtained. The prediction of the quasi-dc electric fields and the plasma density from this model is, in the first-order approximation, reasonably consistent with the CRIT II measurement made inside

a barium stream. However, this model can only apply to the field component perpendicular to the geomagnetic field and therefore cannot explain the parallel component of the quasi-dc electric field which is usually observed in barium release experiments. The maximum ionization yield for the first release in CRIT II was also estimated by this model and found to be consistent with the CRIT II data.

According to the condition in which the Alfvén's critical ionization velocity effect would occur, we found that the space CIV is a self-limiting ionization process in a finite extent barium cloud. This self-limiting ionization mechanism is found to be due to the requirement of the conservation of energy and momentum. As enhanced ionizations take place, the threshold velocity for CIV increases. This could be the reason why the CIV effect was observed in CRIT II but lasted only for a very short period.

One of the most important results we found from this paper is that a high plasma density plasma and a high beam velocity may be the necessary conditions for a high ionization yield. It may explain the high ionization yield in Porcupine [Haerendel, 1982] and CRIT II [Stenbaek-Nielsen *et al.*, 1990b]. The energy transfer efficiency  $\eta_e$  is a crucial parameter which determines the threshold velocity for CIV. However, it is still unclear that how it is dependent on the plasma and the beam parameters. In addition, the current limiting factor  $\alpha$  is also an important parameter which is directly proportional to  $n_a$ . Unfortunately, it is not solved in this paper. A future work on these two parameters will definitely benefit our understanding of the beam release space CIV experiments.

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